

Varshitha Gudimalla

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Education

University at Albany

Masters of Science in Data Science

Albany, New York

May 2025

- **Relevant Coursework:** Advanced Statistics, Machine Learning, Big Data Analytics, Data Mining, Business Intelligence, Statistical Computing.

CMR Institute of Technology, Hyderabad

Bachelors of Technology in Computer Science

India

May 2022

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Software Engineering, Object-Oriented Programming, Web Technologies.

Experience

Data Engineer

FedEx

January 2025 – Present

New Jersey

- Developed scalable ETL pipelines in Databricks using PySpark, Delta Lake, and Python to ingest, clean, and transform high-volume transactional and PII datasets from logistics, customer interaction, and order management systems for downstream analytics and ML model training.
- Orchestrated automated end-to-end data workflows using Apache Airflow, Databricks Workflows, AWS Step Functions, and Jenkins, enabling daily ingestion, transformation, and validation with minimal manual intervention across production environments.
- Integrated key AWS services including S3 (data lake storage), Glue Catalog (metadata management), and Redshift (analytical querying); modularized reusable Python components and parameterized notebook logic for enterprise-scale processing across multi-tenant projects.
- Implemented robust data quality and observability frameworks using Great Expectations, Delta Lake constraints, and schema validation, ensuring integrity across regulatory PII data pipelines and supporting ML teams with trustworthy feature tables for fraud detection, churn modeling, and logistics prediction use cases.
- Collaborated with data scientists, analysts, and DevOps teams to optimize pipeline performance, enforce access control on sensitive data (using IAM roles and Unity Catalog), and enable CI/CD with Git-based version control and release workflows.
- Worked with both batch and near-real-time processing needs; tuned cluster performance, optimized joins and I/O in Spark, and profiled job runtimes to ensure SLA adherence across critical business data domains

Data Engineer

Knowledge Solutions

June 2022 – July 2023

Hyderabad, India

- Worked on developing PySpark-based ETL pipelines in Databricks to transform ~50K weekly customer survey and transactional datasets, enabling feature engineering for NPS prediction, churn analysis, and cost optimization insights.
- Implemented end-to-end orchestration of ingestion workflows using Apache Airflow and AWS DataSync, automating data movement from on-premise sources and APIs into AWS S3 for centralized analytics.
- Developed CI/CD pipelines using Jenkins to deploy and manage Databricks notebooks and job clusters, ensuring consistent delivery across dev, test, and production environments.
- Worked on integrating AWS Glue Catalog and Delta Lake for scalable metadata management and version-controlled data lake architecture supporting analytics and ML models.
- Collaborated with data scientists to prepare ML-ready feature tables, integrating with fraud detection, sentiment analysis, and retention modeling pipelines powered by Amazon Bedrock and SageMaker.
- Built interactive Power BI dashboards to track CSAT, NPS, and churn KPIs; implemented scheduled updates and downstream integration with Salesforce CRM to support real-time customer insights.

Data Engineer

CloudEnd Platform Pvt Ltd

June 2021 – May 2022

Hyderabad, India

- Constructed ETL pipelines using Databricks, AWS Glue, and PySpark to process therapy logs and survey responses into Delta Lake for compliance and analytics.

- Ingested structured and semi-structured data from S3, flat files, and REST APIs, standardizing metadata in Glue Catalog and Hive Metastore.
- Built forecasting pipelines using Python and Prophet in Databricks to predict patient adherence and treatment outcomes, aiding clinical strategy.
- Scheduled and monitored workflows using Apache Airflow DAGs, with deployment automation via Jenkins for nightly refreshes and alerting.
- Wrote optimized SQL procedures and views for patient segmentation and outcome metrics, used by reporting teams in Power BI.
- Maintained documentation, Git version control, and IAM-based access policies aligned with healthcare data governance standards.

Technical Skills

- **Languages:** Python, SQL, PySpark
- **Big Data & Processing:** Apache Spark, Delta Lake, Hive, SparkSQL
- **Cloud & Storage:** AWS (S3, Glue, Redshift, Lambda, Step Functions), Snowflake
- **Workflow Orchestration:** Apache Airflow, Databricks Workflows, Jenkins
- **Data Quality & Governance:** Great Expectations, Glue Catalog, Hive Metastore
- **DevOps & Infra:** Git, Terraform, CI/CD
- **Visualization:** Power BI
- **ML & Forecasting:** Prophet, MLflow, Scikit-learn
- **Messaging & Streaming:** Apache Kafka

Projects

Global Inflation Monitor | *Spark, Snowflake, Tableau, ETL*

- Built end-to-end Spark and Snowflake ETL pipelines and delivered a Tableau dashboard enabling analysts to track real-time inflation and wage data across 190+ countries, reducing reporting delays by 80%.

Customer Churn Prediction | *Python, Machine Learning, Tableau, Predictive Modeling*

- Developed predictive churn models and Tableau dashboards to surface high-risk telecom customers and behavioral patterns, empowering business analysts to drive targeted retention strategies and policy actions.